

Connect and Protect: Networks and Network Security

Module 1

Network architecture

Get started with the course

Introduction to Course 3

Video • 1 min

Course 3 overview

Reading • 4 min

Helpful resources and tips

Reading • 4 min

Introduction to networks

Welcome to module 1

Video • 29 sec

Chris: My path to cybersecurity

Video • 3 min

What are networks?

Video • 1 min

Tina: Working in network security

Video • 2 min

Emmanuel: Useful skills for network security

Video • 2 min

Network tools

Video • 2 min

Network components, devices, and diagrams

Reading • 4 min

Practice: Design a local area network

Ungraded Plugin • 10 min

Cloud networks

Video • 2 min

Cloud computing and software-defined networks

Reading • 8 min

Identify: Functions of network tools

Ungraded Plugin • 10 min

Test your knowledge: Introduction to networks

Practice Assignment • Grade: 100%

Network communication

Introduction to network communication

Video • 2 min

The TCP/IP model

Video • 2 min

The four layers of the TCP/IP model

Video • 1 min

Learn more about the TCP/IP model

Reading • 4 min

The OSI model

Reading • 8 min

Test your knowledge: Network communication

Practice Assignment • Grade: 100%

Local and wide network communication

IP addresses and network communication

Video • 2 min

Components of network layer communication

Reading • 4 min

Test your knowledge: Local and wide network communication

Practice Assignment • Grade: 100%

Review: Network architecture

Wrap-up

Video • 39 sec

Glossary terms from module 1

Reading • 4 min

Module challenge 1

Graded Assignment • Grade: 100%

Module 2

Network operations

Introduction to network protocols

Welcome to module 2

Video • 1 min

Network protocols

Video • 3 min

Common network protocols

Reading • 4 min

Additional network protocols

Reading • 4 min

Arturo: Working in network security

Video • 2 min

Wireless protocols

Video • 1 min

The evolution of wireless security protocols

Reading • 4 min

Test your knowledge: Recognize different network protocols

Practice Assignment • Grade: 100%

System identification

Firewalls and network security measures

Video • 4 min

Virtual private networks (VPNs)

Video • 2 min

Security zones

Video • 3 min

Subnetting and CIDR

Reading • 4 min

Proxy servers

Video • 3 min

Virtual networks and privacy

Reading • 4 min

VPN protocols: Wireguard and IPsec

Reading • 4 min

Test your knowledge: System identification

Practice Assignment • 10 min

Review: Network operations

Wrap-up

Video • 51 sec

Glossary terms from module 2

Reading • 4 min

Module challenge 2

Graded Assignment • 50 min

Module 3

Secure against network intrusions

Module 4

Security hardening

Virtual networks and privacy

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This section of the course covered a lot of information about network operations. You reviewed the fundamentals of network architecture and communication and can now use this knowledge as you learn how to secure networks. Securing a private network requires maintaining the confidentiality of your data and restricting access to authorized users.

In this reading, you will review several network security topics previously covered in the course, including virtual private networks (VPNs), proxy servers, firewalls, and security zones. You'll continue to learn more about these concepts and how they relate to each other as you continue through the course.

Common network protocols

Network protocols are used to direct traffic to the correct device and service depending on the kind of communication being performed by the devices on the network. Protocols are the rules used by all network devices that provide a mutually agreed upon foundation for how to transfer data across a network.

There are three main categories of network protocols: communication protocols, management protocols, and security protocols.

1. Communication protocols are used to establish connections between servers. Examples include TCP, UDP, and Simple Mail Transfer Protocol (SMTP), which provides a framework for email communication.

2. Management protocols are used to troubleshoot network issues. One example is the Internet Control Message Protocol (ICMP).

3. Security protocols provide encryption for data in transit. Examples include IPsec and SSL/TLS.

Some other commonly used protocols are:

• HyperText Transfer Protocol (HTTP). HTTP is an application layer communication protocol. This allows the browser and the web server to communicate with one another.

• Domain Name System (DNS). DNS is an application layer protocol that translates, or maps, host names to IP addresses.

• Address Resolution Protocol (ARP). ARP is a network layer communication protocol that maps IP addresses to physical machines or a MAC address recognized on the local area network.

Wi-Fi

This section of the course also introduced various wireless security protocols, including WEP, WPA, WPA2, and WPA3. WPA3 encrypts traffic with the Advanced Encryption Standard (AES) cipher as it travels from your device to the wireless access point. WPA2 and WPA3 offer two modes: personal and enterprise. Personal mode is best suited for home networks while enterprise mode is generally utilized for business networks and applications.

Network security tools and practices

Firewalls

Previously, you learned that firewalls are network virtual appliances (NVA) or hardware devices that inspect and can filter network traffic before it's permitted to enter the private network. Traditional firewalls are configured with rules that tell it what types of data packets are allowed based on the port number and IP address of the data packet.

There are two main categories of firewalls.

• **Stateless:** A class of firewall that operates based on predefined rules and does not keep track of information from data packets

• **Stateful:** A class of firewall that keeps track of information passing through it and proactively filters out threats. Unlike stateless firewalls, which require rules to be configured in two directions, a stateful firewall only requires a rule in one direction. This is because it uses a "state table" to track connections, so it can match return traffic to an existing session

Next generation firewalls (NGFWs) are the most technologically advanced firewall protection. They exceed the security offered by stateful firewalls because they include deep packet inspection (a kind of packet sniffing that examines data packets and takes actions if threats exist) and intrusion prevention features that detect security threats and notify firewall administrators. NGFWs can inspect traffic at the application layer of the TCP/IP model and are typically application aware. Unlike traditional firewalls that block traffic based on IP address and ports, NGFWs rules can be configured to block or allow traffic based on the application. Some NGFWs have additional features like Malware Sandboxing, Network Anti-Virus, and URL and DNS Filtering.

Proxy servers

A proxy server is another way to add security to your private network. Proxy servers utilize network address translation (NAT) to serve as a barrier between clients on the network and external threats. Forward proxies handle queries from internal clients when they access resources external to the network. Reverse proxies function opposite of forward proxies; they handle requests from external systems to services on the internal network. Some proxy servers can also be configured with rules, like a firewall. For example, you can create filters to block websites identified as containing malware.

Virtual Private Networks (VPN)

A VPN is a service that encrypts data in transit and disguises your IP address. VPNs use a process called encapsulation. Encapsulation wraps your unencrypted data in an encrypted data packet, which allows your data to be sent across the public network while remaining anonymous. Enterprises and other organizations use VPNs to help protect communications from users' devices to corporate resources. Some of these resources include servers or virtual machines that host business applications. Individuals also use VPNs to increase personal privacy. VPNs protect user privacy by concealing personal information, including IP addresses, from external servers. A reputable VPN also minimizes its own access to user internet activity by using strong encryption and other security measures. Organizations are increasingly using a combination of VPN and SD-WAN capabilities to secure their networks. A software-defined wide area network (SD-WAN) is a virtual WAN service that allows organizations to securely connect users to applications across multiple locations and over large geographical distances.

Key takeaways

There are three main categories of network protocols: communication, management, and security protocols. In this reading, you learned the fundamentals of firewalls, proxy servers, and VPNs. More organizations are implementing a cloud-based approach to network security by incorporating a combination of VPN and SD-WAN capabilities as a service.

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This reading provides a review of network security topics, including network protocols, wireless security, firewalls, proxy servers, and Virtual Private Networks (VPNs).

Network Protocols

• Network protocols are rules that direct traffic and facilitate data transfer across a network, categorized into communication, management, and security protocols.

• Examples include TCP, UDP, SMTP for communication; ICMP for management; and IPsec, SSL/TLS for security, along with HTTP, DNS, and ARP.

Wireless Security Protocols

• Various wireless security protocols like WEP, WPA, WPA2, and WPA3 exist, with WPA3 offering the strongest encryption using AES.

• WPA2 and WPA3 offer personal mode for home networks and enterprise mode for business networks.

Network Security Tools and Practices

• Firewalls inspect and filter network traffic, with stateless firewalls using predefined rules and stateful firewalls tracking connections for proactive threat filtering.

• Next-generation firewalls (NGFWs) offer advanced protection through deep packet inspection, intrusion prevention, and application-aware traffic control.

Proxy Servers and VPNs

• Proxy servers act as a barrier between internal clients and external threats, using Network Address Translation (NAT) and configurable rules to filter traffic.

• VPNs encrypt data in transit and disguise IP addresses through encapsulation, protecting communications for both organizations and individuals, often combined with SD-WAN for enhanced security.

And if you want to continue exploring this topic, try one of these follow-up questions:

What is the difference between stateless and stateful firewalls?

How can you configure a proxy server to block malicious websites?

How would combining VPNs and SD-WAN improve network security?

Ask me anything

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